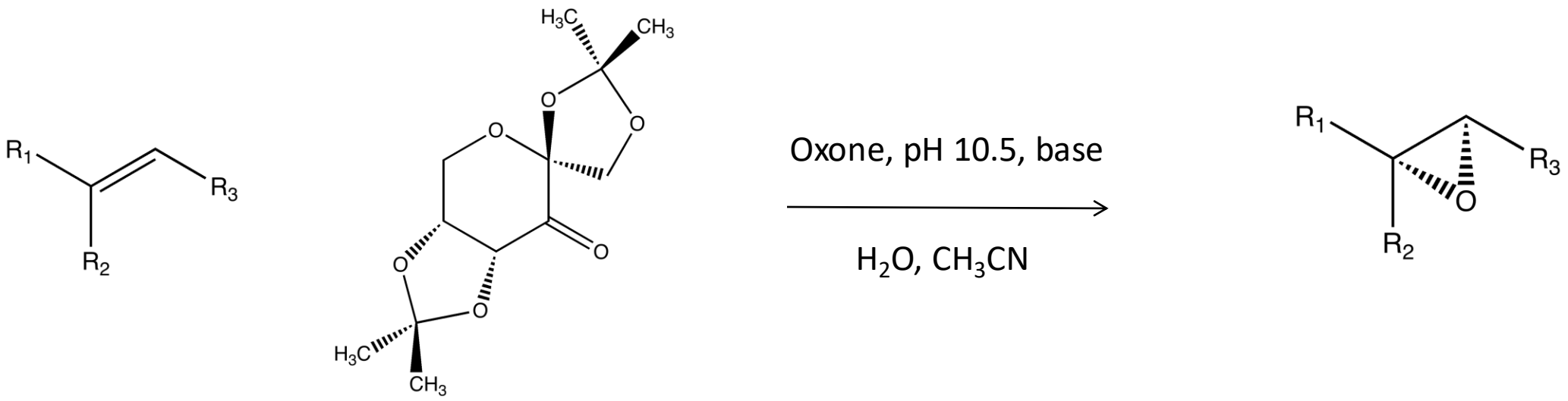
Isotope Effects and the Nature of Enantioselectivity in the Shi Epoxidation. The Importance of Asynchronicity

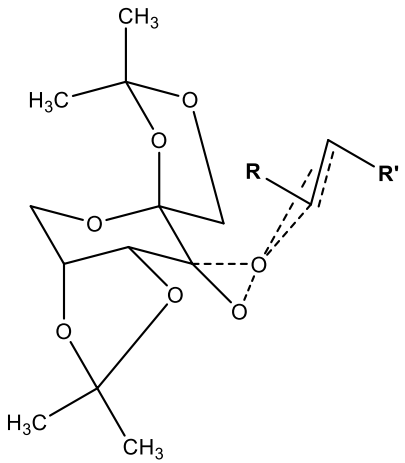
Andrew Ross

General Shi Epoxidation



The Shi Epoxidation - Enantioselective Considerations

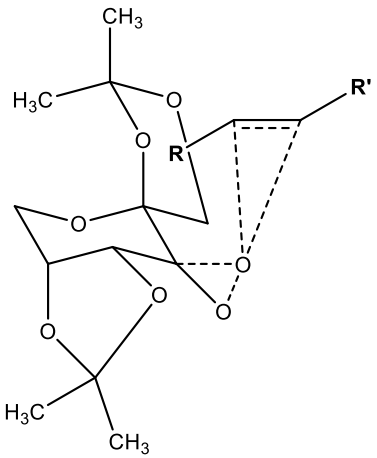
Spiro T.S.



Trans-disubstitued alkene approaches the dioxirane to minimize steric interactions with the beta-axial dioxolane ring (positioned adjacent to the heterocyclic oxygen atom).

Produces Observed Major Products

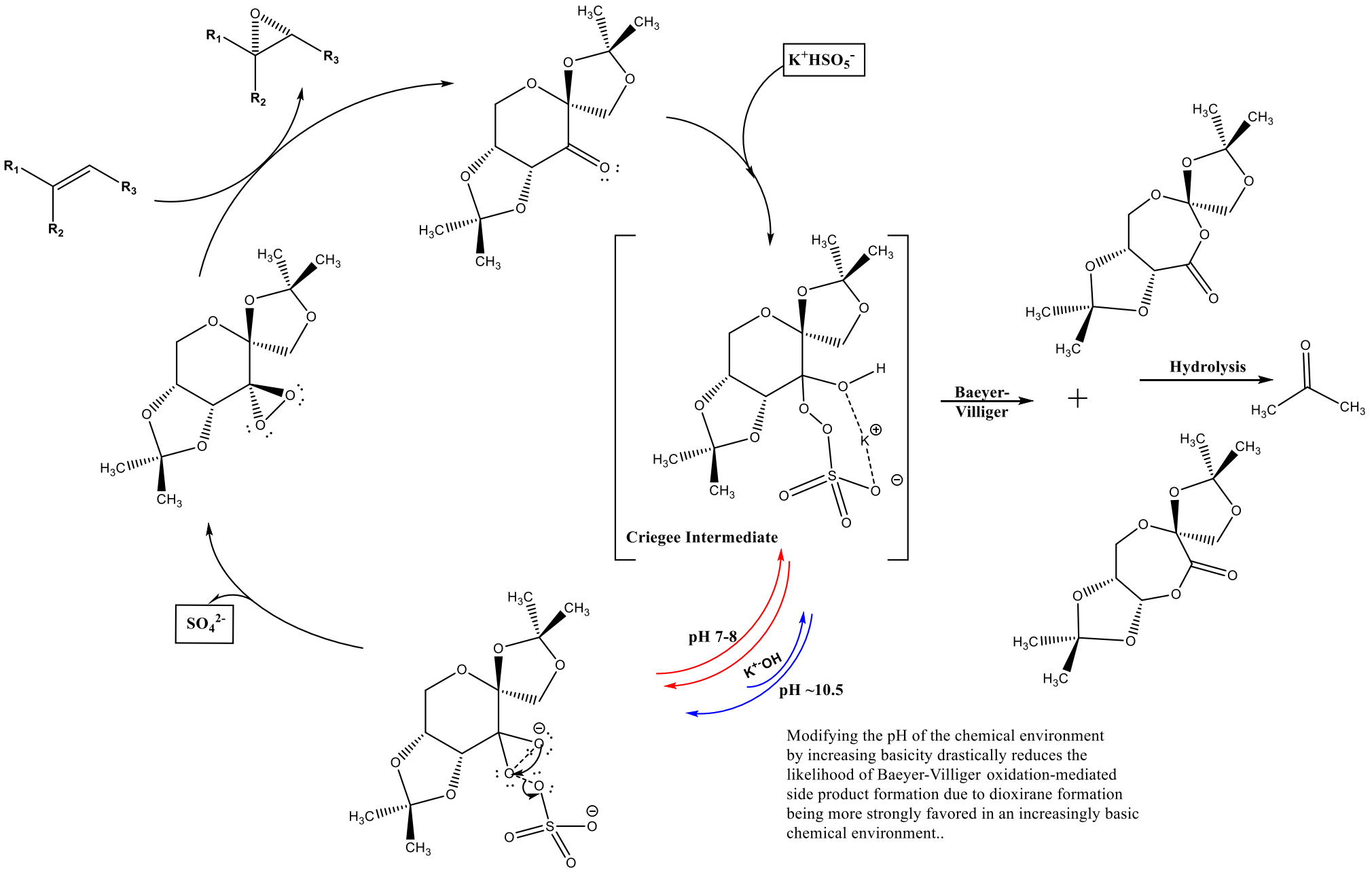
Planar T.S.



Trans-disubstitued alkene approaches the dioxirane to minimize steric interactions with the beta-axial dioxolane ring (positioned adjacent to the heterocyclic oxygen atom).

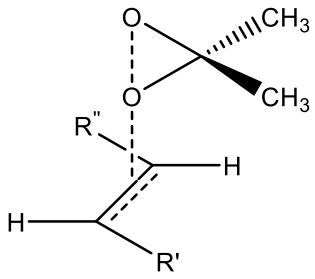
Produces Observed Minor Products

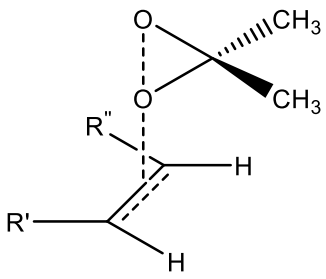
***Asynchronous T.S.*:** the preference for the Spiro T.S. stems from there being differential heterocyclic bond formation of the incipient C-O bonds (this is a highly asynchronous process).

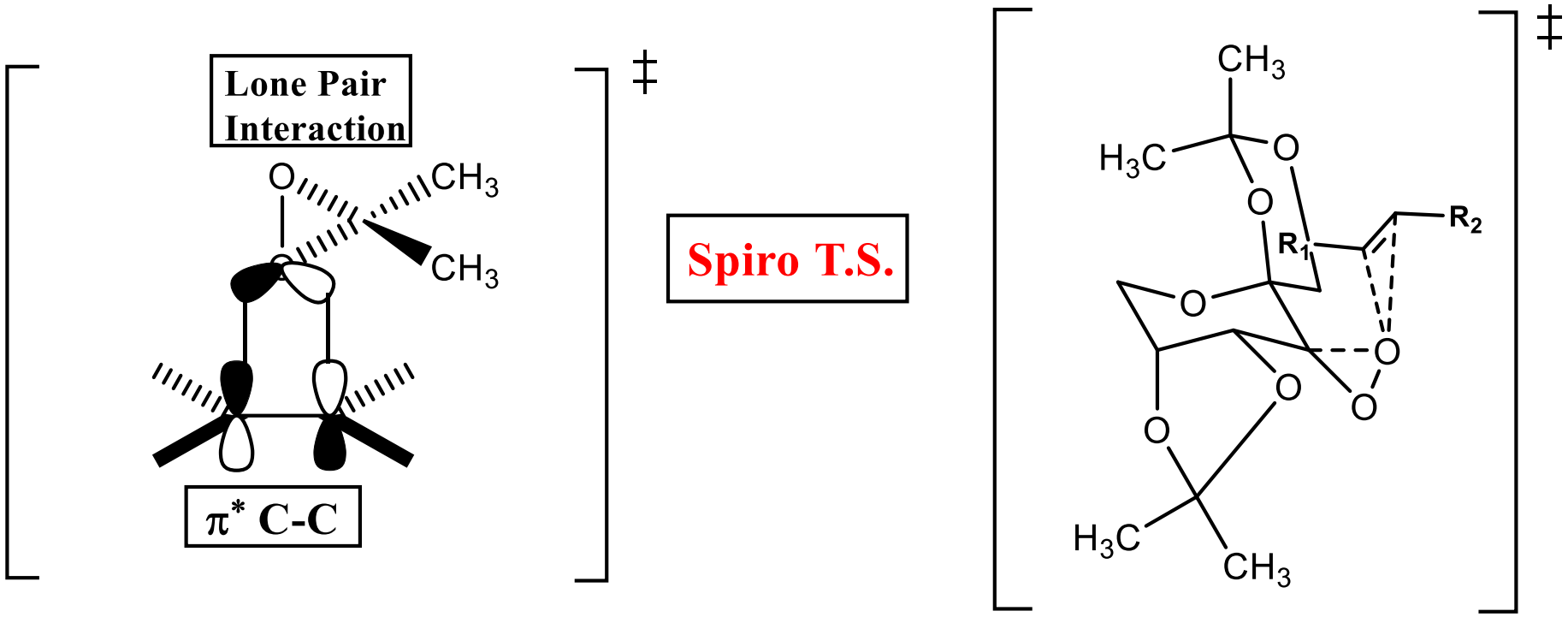


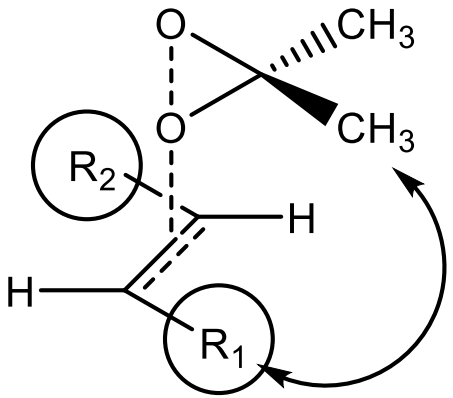
Modifying the pH of the chemical environment by increasing basicity drastically reduces the likelihood of Baeyer-Villiger oxidation-mediated side product formation due to dioxirane formation being more strongly favored in an increasingly basic chemical environment..

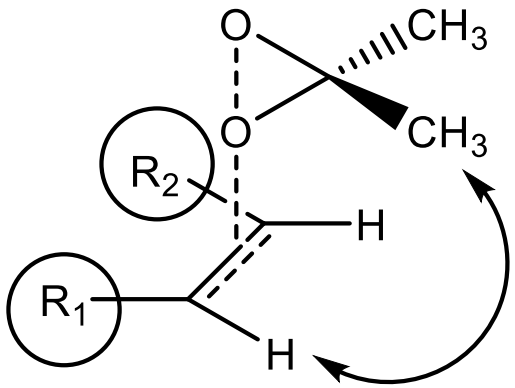
Transition State Considerations: RLS

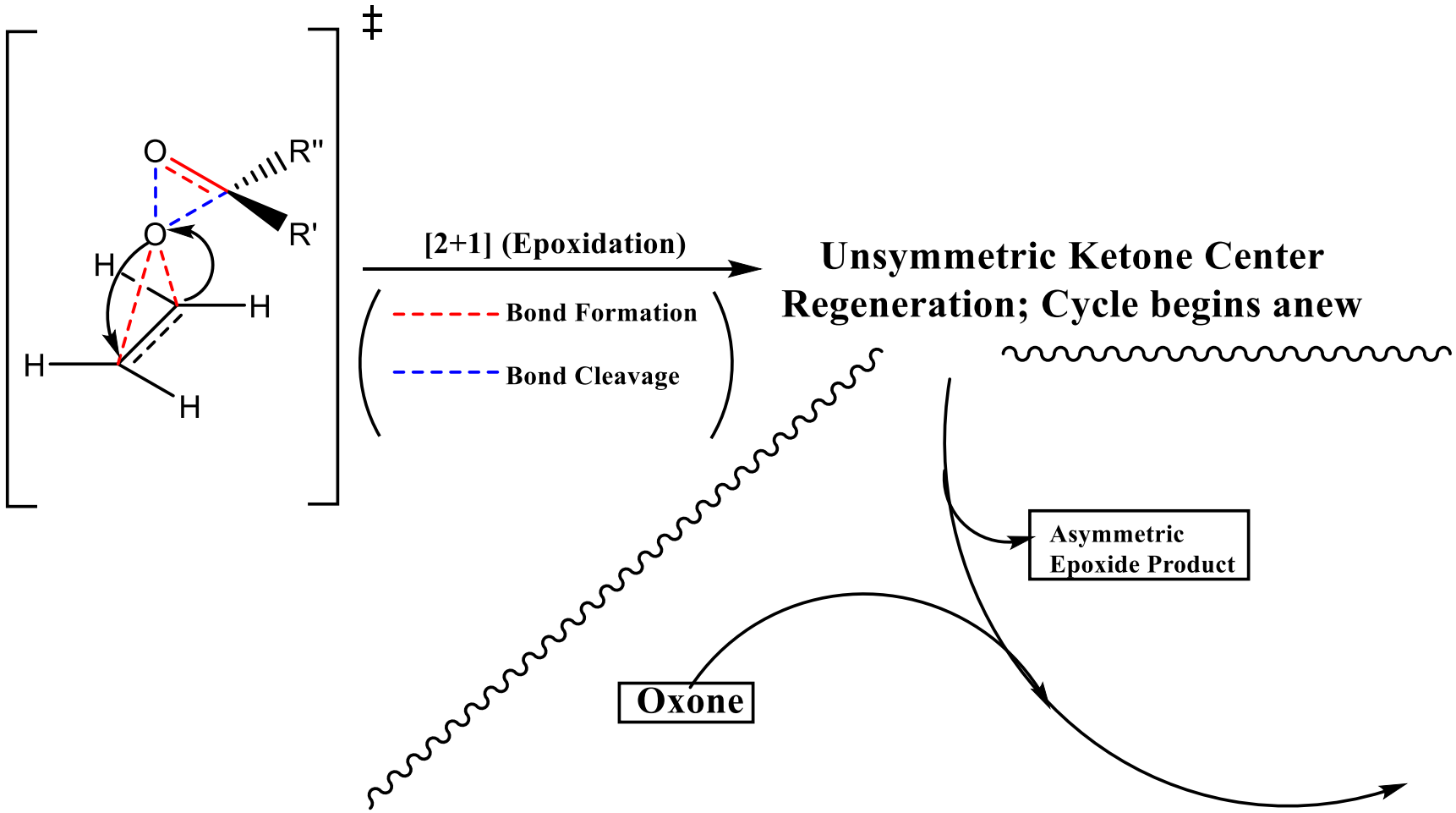




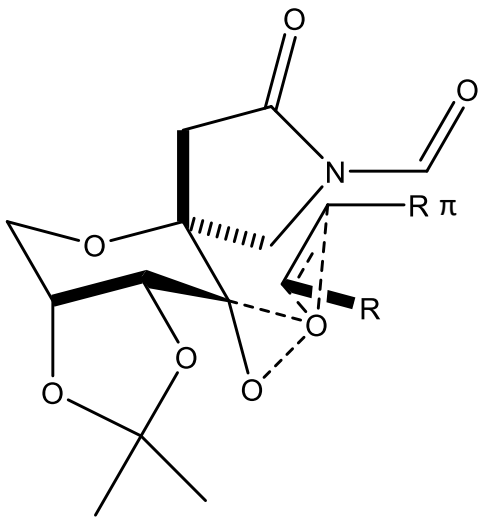

Lone Pair Interaction
O
CH3
CH3
π* C-C
‡
Spiro T.S.
CH3
H3C
O
O
R1
R2
O
O
O
O
O
H3C
CH3
‡


O
CH3
O
CH3
R2
H
H
R1

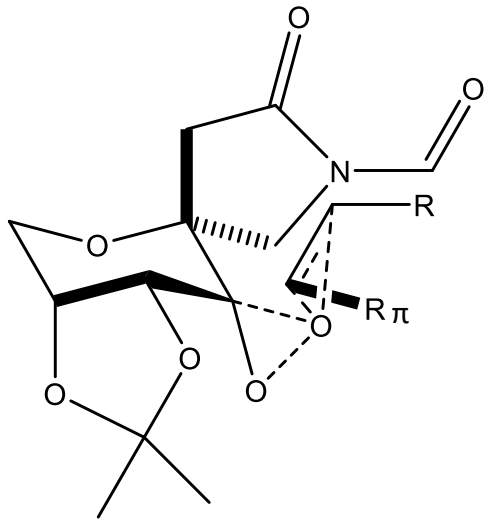

O
CH3
O
CH3
R2
H
R1
H


‡
O
R''
R'
O
H
H
H
H
[2+1] (Epoxidation)
Bond Formation
Bond Cleavage
Unsymmetric Ketone Center
Regeneration; Cycle begins anew
Asymmetric
Epoxide Product
Oxone

Cis-Alkenes and Terminal Alkenes



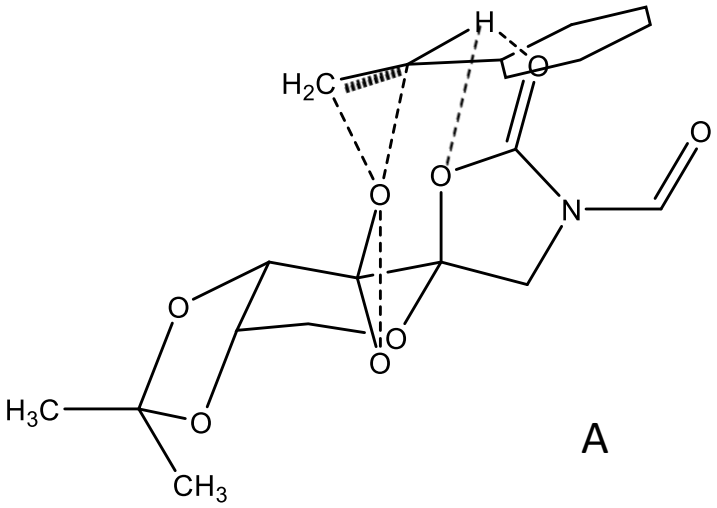
vs



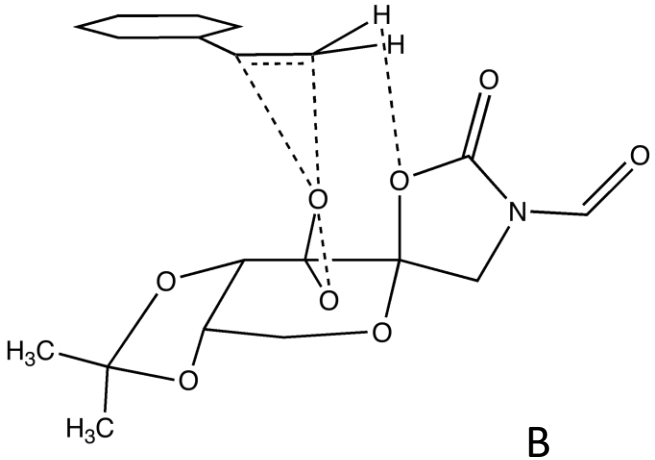
The π conjugating system prefers to be proximal to the oxazolidinone ring.

Lowest Energy Transition Structures

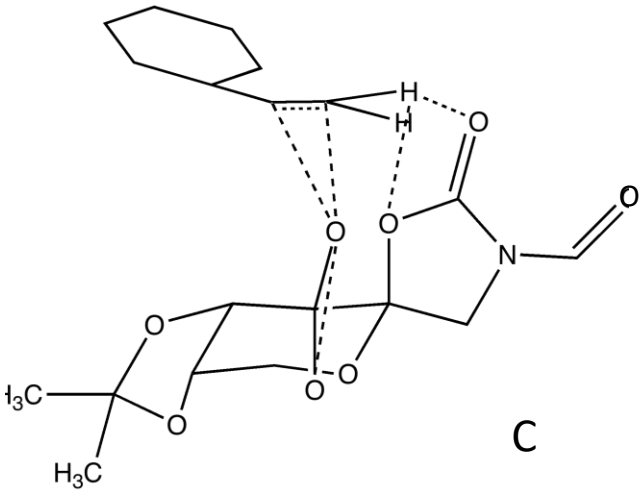
Transition Structure	Energy reaction barriers (kcal/mol)
A	5.9
B	7.4
C	7.1
D	7.3



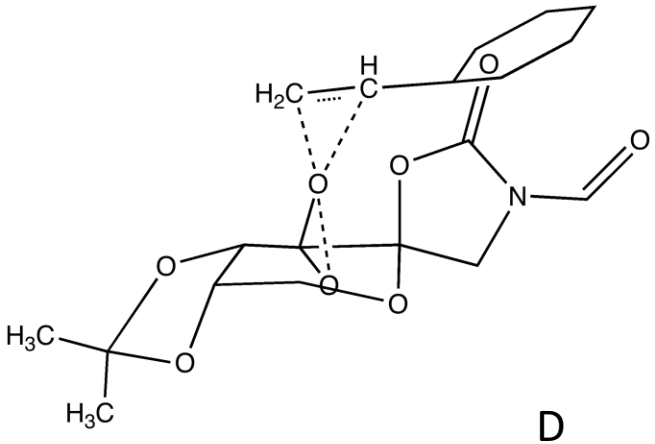
A



B

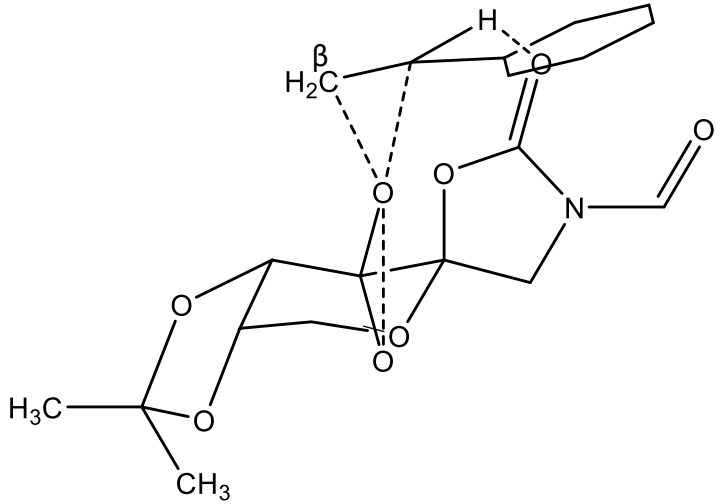


C



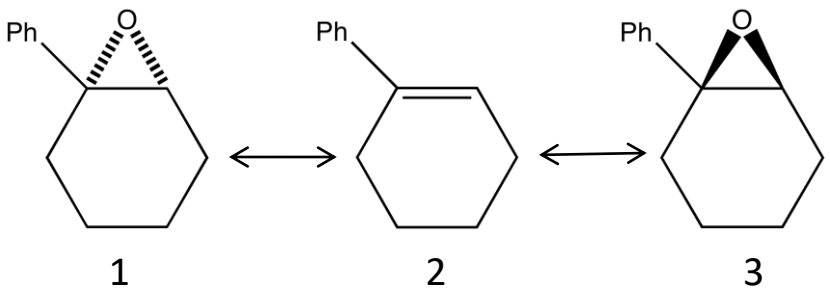
D

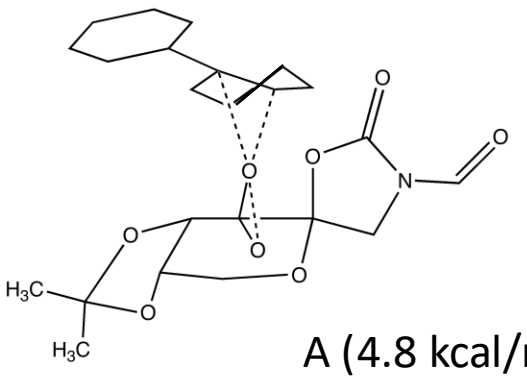
Lowest energy Transition Structure



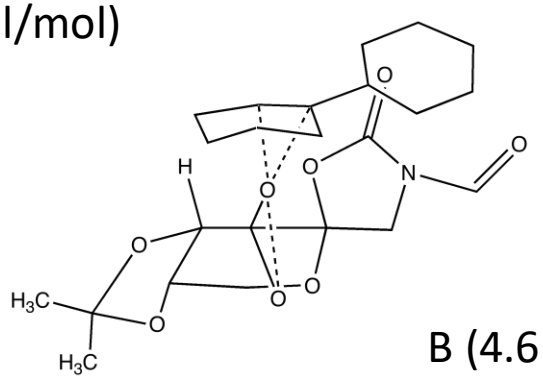
- Preference for this transition state is due to having the olfinic hydrogen in a favorable position.
- Favorable C_β--O—O bond angle in this transition state.
- Charge stabilization of sterene by oxazolidinone ring

1-phenyl-cyclohexene



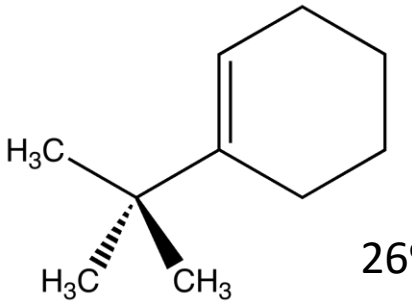


A (4.8 kcal/mol)

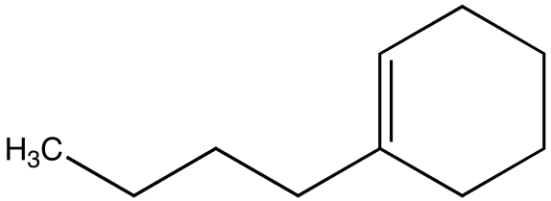


B (4.6 kcal/mol)

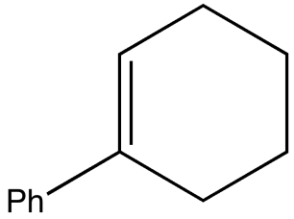
Effect of a smaller R_1



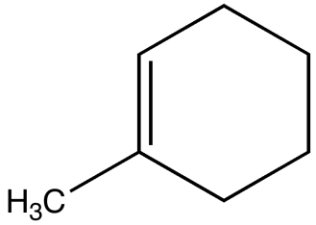
26% ee



79% ee

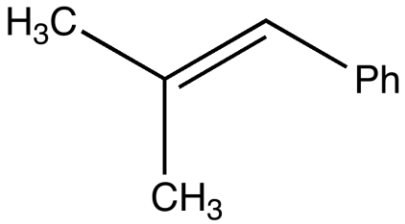


81% ee

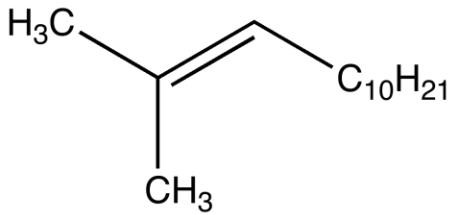


98% ee

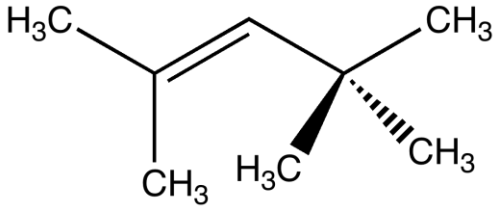
Effect of larger R_3



76% ee

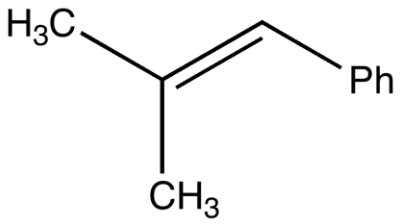


86% ee

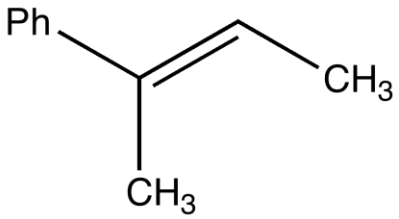


91% ee

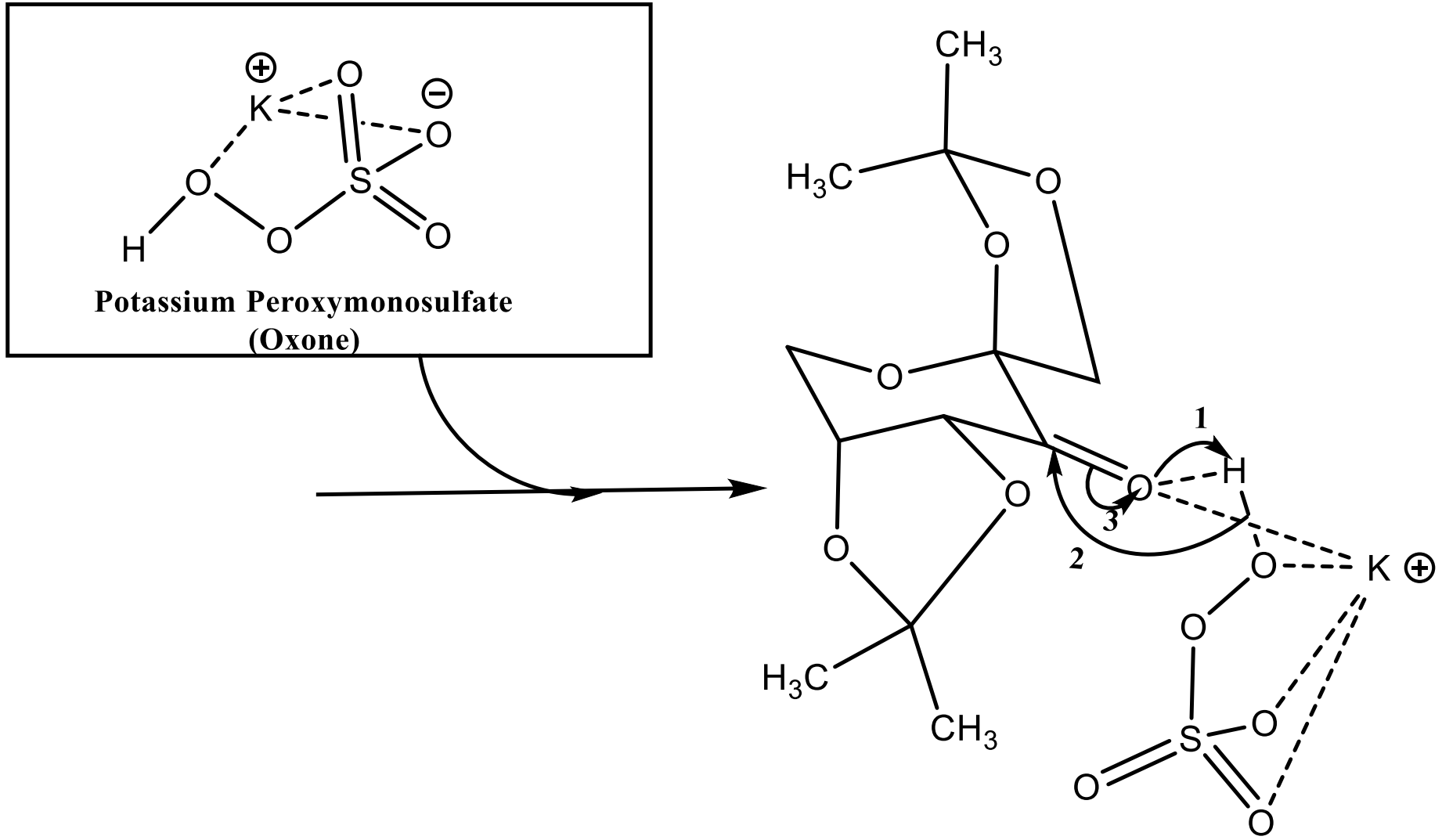
Comparing the size of R_1 and R_3

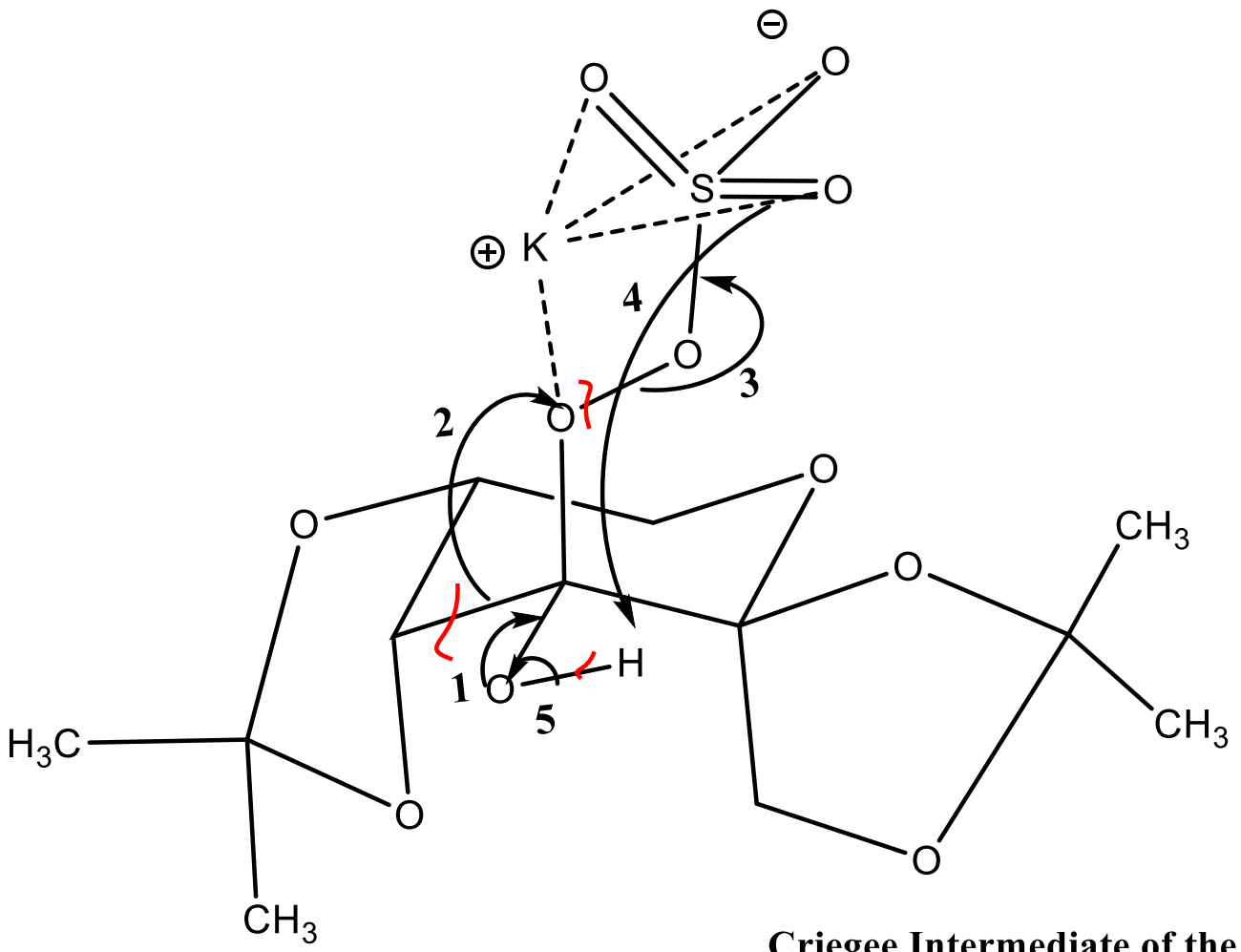


76% ee

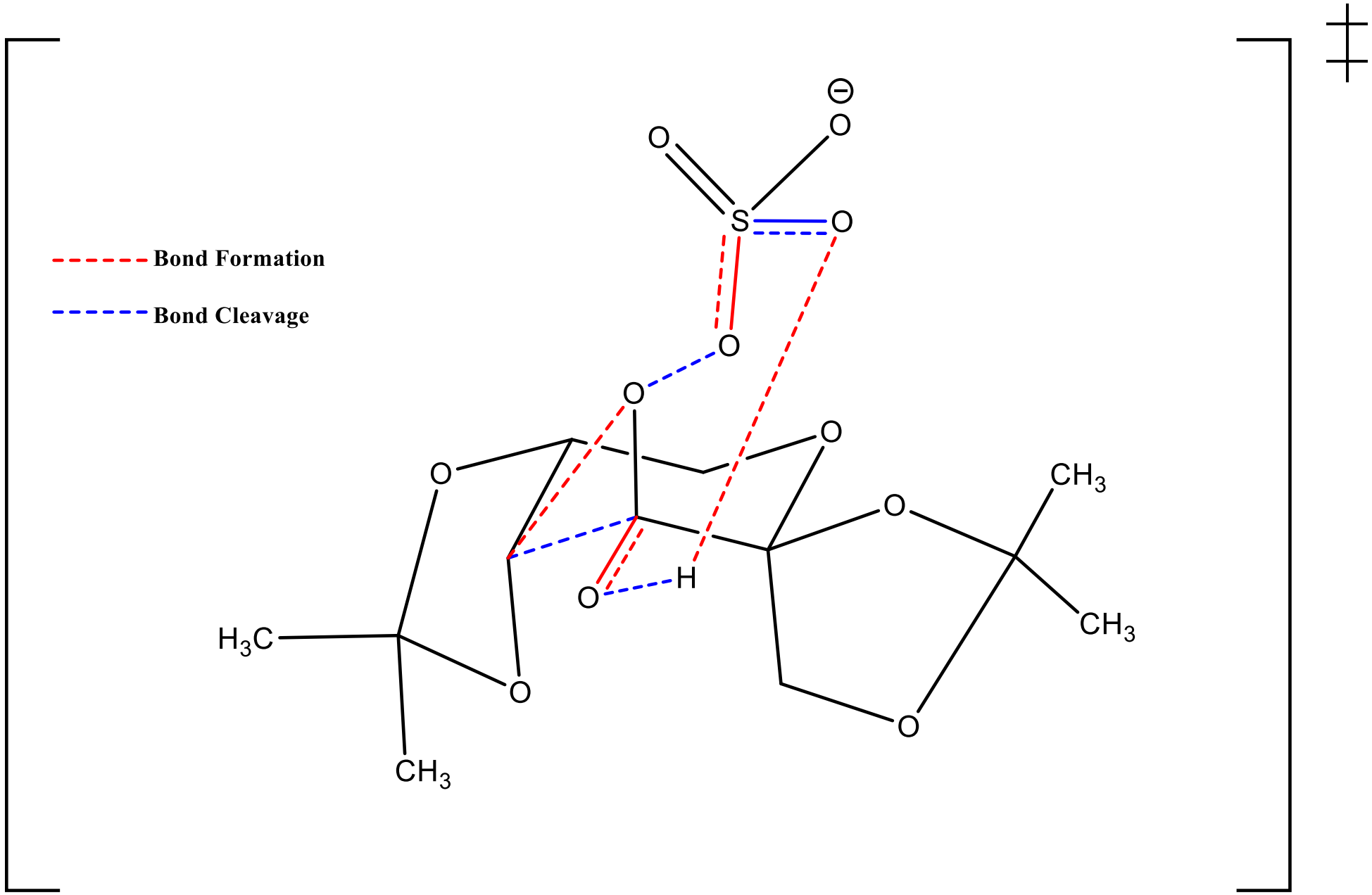


97%ee

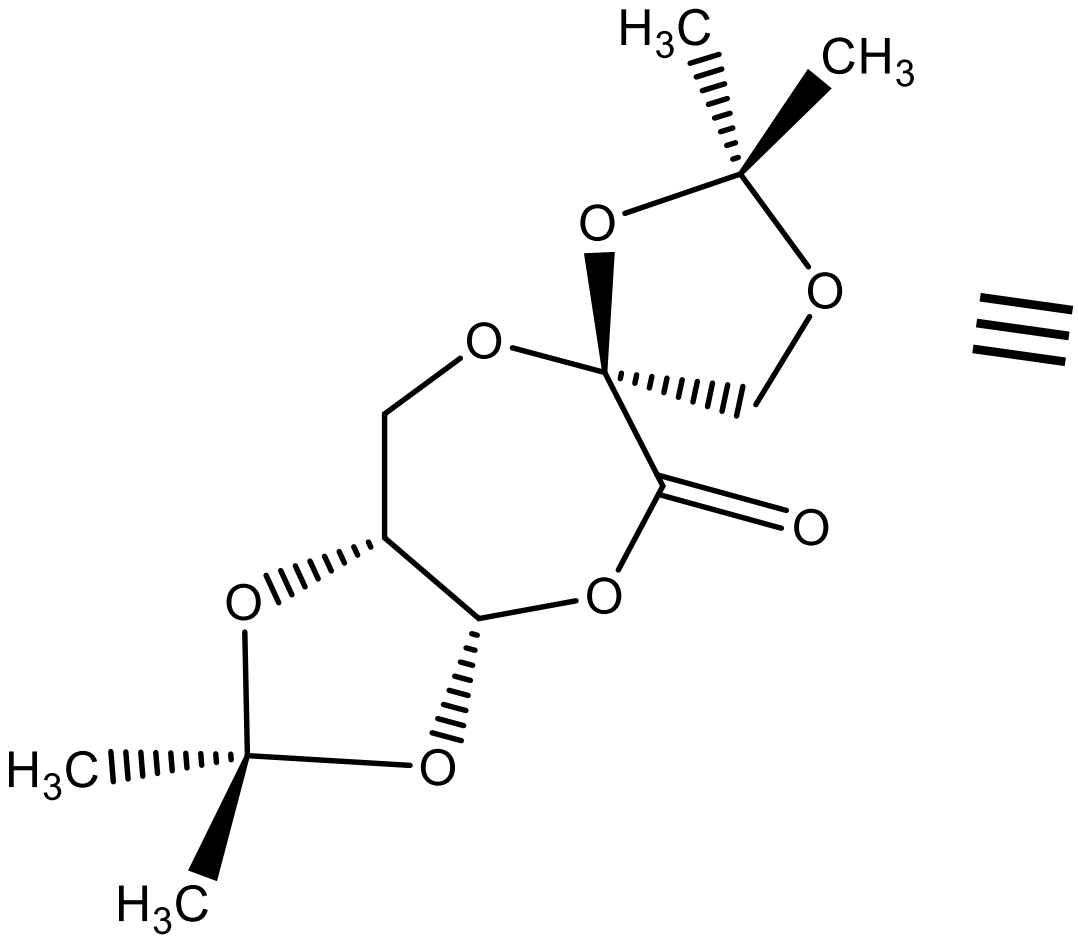

Potassium Peroxymonosulfate
(Oxone)
1
2
3



Criegee Intermediate of the electrophilically mediated cycloexpansion reaction.

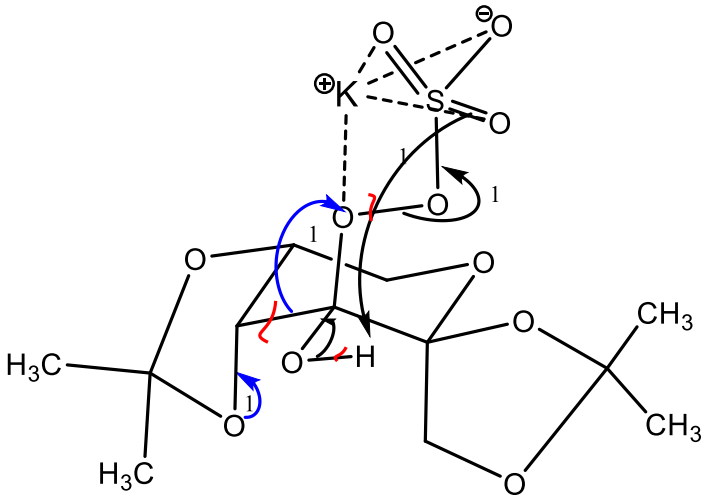

Bond Formation
Bond Cleavage
O
O
S
O
O
O
O
O
O
O
H
O
H_3C
CH_3
CH_3
CH_3
O
O
‡

1 of 2 cycloaddition Byproducts from the pH=sensitive BV-mediated cycloexpansion reaction



The Baeyer-Villiger Oxidation – A Brief Review

For unsymmetrical ketones, the structure depends on which group migrates. This is known as its *migratory aptitude*. Migratory aptitude is directly proportional to the proficiency with which any given group is able to stabilize a formal positive charge



Is Neighboring group participation possible?

References

Frohn M; Shi Y: Chiral Ketone-Catalyzed Asymmetric Epoxidation of Olefins. *Synthesis* **2000,** *14*, 1979−2000.

Grein F; Chen AC; Edwards D; Crudden CM: Theoretical and Experimental studies on the Baeyer-Villiger Oxidation of Ketones and the Effect of α-Halo Substituents. *J. Org. Chem.* **2006,** *71*, 861−872.

Singleton DA; Wang Z: Isotope Effects and the Nature of Enantioselectivity in the Shi Epoxidation. The Importance of Asynchronicity. *J. Am. Chem. Soc.* **2005,** *127*, 6679−6685.

Wang B; Wong OA; Zhao M; Shi Y: Asymmetric Epoxidation of 1,1-Disubstituted Terminal Olefins by Chiral Dioxirane via a Planar-like Transition State. *J. Org. Chem.* **2008,** *73*, 9539−9543.

Wang Z; Tu Y; Frohn M; Zhang JR; Shi Y: An Efficient Catalytic Asymmetric Epoxidation Method. *J. Am. Chem. Soc.* **1997,** *119*, 11224−11235.